United States Patent	12396528
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Carroll; Taishah Amiyra

Tamper-proof wearable device and system thereof

Abstract

A device configured to be worn by a user is described. The device comprises a first member having an opening and a second member configured to be detachably engaged with the first member. The device further comprises a chamber comprising liquid ink, the liquid ink being in fluid communication with the opening via a passage extending between the chamber and the opening. The device further comprises a locking mechanism configured to switch between a locked state and an unlocked state, and an ejection module configured to eject the liquid ink through the opening of the first member when the locking mechanism is at the locked state thereof.

Inventors: **Carroll; Taishah Amiyra (Clifton Heights, PA)**

Applicant: **Carroll; Taishah Amiyra** (Clifton Heights, PA)

Family ID: **1000007186139**

Assignee: **Carroll; Taishah A. (Clifton Heights, PA); Kamala; Ari Blyss (Clifton Heights, PA)**

Appl. No.: **18/204830**

Filed: **June 01, 2023**

Related U.S. Application Data

us-provisional-application US 63348202 20220602

Publication Classification

Int. Cl.: **E05B73/00** (20060101); **A44C7/00** (20060101); **B65D51/00** (20060101); **B65D81/32** (20060101)

U.S. Cl.:

CPC **A44C7/001** (20130101); **B65D51/002** (20130101); **B65D81/3205** (20130101); **E05B73/0017** (20130101);

Field of Classification Search

CPC: G08B (13/2434); G08B (15/02); G08B (21/0269); G08B (21/0288); G08B (13/19684); G08B (13/2445); G08B (15/004); G08B (21/0202); G08B (21/0205); G08B (21/0252); G08B (21/0261); G08B (21/0275); G08B (21/028); G08B (21/0283); G08B (21/0291); G08B (25/016); E05B (73/0017); E05B (47/0004); E05B (47/0005); E05B (39/002); E05B (15/04); E05B (51/005); E05B (47/0002); Y10T (436/25); H04L (2209/80); H04L (2209/84); H04L (2209/88); H04L (63/0428); H04L (63/101); H04L (9/3226); G16H (10/60); G16H (10/65); G06Q (40/08); G06K (19/06037); G06K (19/0776); G06K (7/1417); G01S (5/0072); F41H (9/10); A41D (1/002); H04W (4/029); H04W (12/00); H04W (12/02); H04W (12/08); H04W (12/77); H04W (48/08); H04W (48/16); H04W (64/00); A44C (7/00); G07G (3/003); G07F (19/205); G01N (2001/007); E05G (1/12); E05G (5/00); B01L (3/502); A61B (10/007); A61B (2010/0009); G07D (11/125); G07D (11/225); G07D (11/40); G07D (9/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6277646	12/2000	Guirguis	422/417	B01L 3/502
7414535	12/2007	Hanabusa et al.	N/A	N/A
2007/0074043	12/2006	Lacey	N/A	N/A
2007/0171047	12/2006	Goodman et al.	N/A	N/A
2009/0058659	12/2008	Handyside	340/572.9	G08B 15/02
2014/0118149	12/2013	Elias	N/A	N/A
2017/0132899	12/2016	Theocharous	N/A	N/A
2019/0114887	12/2018	Brantley	N/A	N/A
2020/0347647	12/2019	Lindholm	N/A	E05B 15/04
2021/0297813	12/2020	Camp	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
9308712	12/1992	WO	N/A

Primary Examiner: Pendleton; Dionne

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application 63/348,202, filed Jun. 2, 2022, which is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION
(1) The present invention relates generally to wearable devices, and more particularly to a tamper-proof tracking device that is worn by a user.

BACKGROUND OF THE INVENTION

(2) Conventionally, devices to keep a track of human movement are known. The devices can be carried by a person, such as an adult or a child. Tracking devices may be bulky and inconvenient to carry all the time. Moreover, a person may misplace the tracking devices and it becomes impossible to track the person from a remote location. Some tracking devices can be attached to clothes or accessories of a person; however, such tracking devices are not tamper-proof. Any unauthorized user can mishandle the tracking devices thereby nullifying the purpose of the tracking devices.

(3) These drawbacks in tracking devices are extremely significant in cases of human trafficking. It is a challenge nowadays for officials and even general public to identify victims and/or culprits of human trafficking. With known devices, officials and guardians of a person can at most track the movement of the person. However, in case a person is abducted and being taken to other places, such as via an airport, the known devices do not allow general public and officials present at the airport to easily identify the victims and/or culprits. Moreover, the culprits can easily deactivate the known devices which further increases the difficulty to prevent human trafficking.

(4) Accordingly, there is an established need for a solution to the problems mentioned above. For instance, there is an established need for a device that is tamper-proof such that the device cannot be deactivated by an unauthorized user. Further, there is an established need for a device that allows onlookers to identify the victims and/or culprits. Further, there is an established need for a device that can be easily and conveniently worn by a user. Further, there is an established need for a device that can be controlled by the user for locking and unlocking purposes, and additionally, can be tracked by a remote user.

SUMMARY OF THE INVENTION

(5) The present invention relates to a wearable device configured to be worn by a user. The device comprises a first member having an opening and a second member configured to be detachably engaged with the first member. The device further comprises a chamber comprising liquid ink, the liquid ink being in fluid communication with the opening via a passage extending between the chamber and the opening. The device further comprises a locking mechanism configured to switch between a locked state and an unlocked state; and an ejection module configured to eject the liquid ink through the opening of the first member when the locking mechanism is at the locked state thereof. Thus, the wearable device the inhibits episodes of tampering when worn by an individual.

(6) In an aspect, when the locking mechanism is at the locked state, the first member is prevented from disengaging with the second member. In an aspect, when the locking mechanism is at the unlocked state, the first member is allowed to disengage from the second member.

(7) In an aspect, the present invention relates to a system comprising the wearable device and a user device communicatively coupled to the wearable device. The wearable device comprises a tracking module and the user device is configured to receive location information of the wearable device from the tracking module.

(8) In an aspect, the system further comprises a key fob communicatively coupled to the wearable device, the key fob being configured to control switching of the locking mechanism between the locked state and the unlocked state.

(9) In an aspect, the present invention relates to a kit comprising the wearable device and the key fob, the kit further comprising a QR code to facilitate association between the key fob and the wearable device.

(10) In an aspect, the wearable device is a piece of jewelry.

(11) In an aspect, the wearable device is an ear ring. In such an aspect, the first member is an ornamental piece of the ear ring, and the second member is a stopper of the ear ring. In an aspect, the ornamental piece is a stud.

(12) In another aspect, the ejection module is operatively connected to the locking mechanism.

(13) In another aspect, the ejection module is activated when the locking mechanism is in the locked state thereof and deactivated when the locking mechanism is in the unlocked state thereof.

(14) In another aspect, the wearable device further comprises a tracking module configured to transmit a location of the wearable device.
(15) In another aspect the tracking module is positioned within either the first member or the second member.
(16) In another aspect, the ejection module is positioned within either the first member or the second member.
(17) In another aspect, the tracking module is a Global Positioning System (GPS) module.
(18) In another aspect, the tracking module comprises a transceiver configured to receive satellite signals and transmit location data indicative of the location of the wearable device; and a processing unit configured for analyzing the satellite signals received by the transceiver and determining the location of the wearable device.
(19) In another aspect, the liquid ink is a permanent ink.
(20) In another aspect, the chamber further comprises a sealing element configured to prevent the liquid ink contained within the chamber from leaking to the passage.
(21) In another aspect, the sealing element is configured to rupture when a predetermined force is exerted thereon.
(22) In another aspect, the sealing element is configured to have a sealed state in which fluid communication between the chamber and the liquid ink contained therein is prevented, and a ruptured state in which fluid communication between the chamber and the liquid ink contained therein is allowed.
(23) In another aspect the ejection module comprises a battery and an ejector coupled to the battery, wherein the ejector is configured to provide the liquid ink with sufficient energy to rupture the sealing element thereby causing a transitioning of the sealing element from the scaled state to the ruptured state.
(24) In another aspect, the ejection module is configured to eject the liquid ink in a 360 degree pattern.
(25) In another aspect, the wearable device is an ear ring.
(26) These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the embodiments and examples, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:
(2) FIG. **1** illustrates a perspective view of a wearable device in an assembled state, in accordance with one embodiment of the present specification;
(3) FIG. **2** illustrates a perspective view of the wearable device in a disassembled state, in accordance with one embodiment of the present specification;
(4) FIG. **3** illustrates a cross-sectional view of a first member of the wearable device, in accordance with one embodiment of the present specification;
(5) FIG. **4** illustrates a system comprising the wearable device, a user device, and a key fob, in accordance with one embodiment of the present specification;
(6) FIG. **5** illustrates the user device in communication with a tracking module of the wearable device, in accordance with one embodiment of the present specification;
(7) FIG. **6** illustrates the wearable device being placed into a pierced earlobe of a user, in accordance with one embodiment of the present specification;

(8) FIG. **7** illustrates the wearable device in place in the earlobe of the user, in accordance with one embodiment of the present specification; and
(9) FIG. **8** illustrates the wearable device with ink being ejected out therefrom, in accordance with one embodiment of the present specification.
(10) Like reference numerals refer to like parts throughout the several views of the drawings.
DETAILED DESCRIPTION
(11) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other embodiments. All of the embodiments described below are exemplary embodiments provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in the drawings. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
(12) In the following description, certain specific details are set forth in order to provide a thorough understanding of various disclosed embodiments. However, one skilled in the relevant art will recognize that embodiments may be practiced without one or more of these specific details, or with other methods, components, materials, and the like. In other instances, well-known elements associated with wearable devices have not been shown or described in detail to avoid unnecessarily obscuring descriptions of the embodiments.
(13) Unless the context requires otherwise, throughout the specification and claims which follow, the word “comprise” and variations thereof, such as, “comprises” and “comprising” are to be construed in an open, inclusive sense, that is, as “including, but not limited to.”
(14) As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise, and the vice versa. It should also be noted that the term “or” is generally employed in its broadest sense, that is, as meaning “and/or” unless the content clearly dictates otherwise.
(15) The headings and Abstract of the Disclosure provided herein are for convenience only and do not interpret the scope or meaning of the embodiments.
(16) Reference is initially made to FIGS. **1-2** in which FIG. **1** illustrates a perspective view of a wearable device **100** configured to be worn by a user, the wearable device **100** being in an assembled state, and FIG. **2** illustrates another perspective view of the wearable device **100**, the wearable device **100** being in a disassembled state.
(17) The term ‘wearable device’ as used in the present specification and claims is intended to refer to various types of devices that may be worn by a user directly (to skin) or indirectly (to fabrics and outfits). In an embodiment, the wearable device **100** is an ear ring that is worn by a user in one or both ears. The term ‘user’ as used in the present specification and claims refers to any person wearing the wearable device **100**, such as a child, a teenager, or an adult.
(18) In the assembled state, the wearable device **100** comprises a first member **110** and a second member **120** coupled to the first member **110**. The first and second members **110**, **120** can be seen as separated in the disassembled state of the wearable device **100** shown in FIG. **2**. The first

member **110** may be in the form of a male connector while the second member **120** may be in the form of a female connector. The wearable device **100** is thus configured to be shifted from the disassembled state to the assembled state, and selectively from the assembled state to the disassembled state.

(19) In the illustrated embodiment, the first member **110** comprises a housing **112** and a protrusion **114** extending from the housing **112**. The first member **110** is configured to be received within the second member **120**, in that, the protrusion **114** is configured to be in engagement with the second member **120**, thereby coupling the first member **110** to the second member **120**. As seen in FIG. **2**, the protrusion **114** may comprise external threading **114**A configured to facilitate coupling of the protrusion **114**, and thus the first member **110**, to the second member **120**. In an embodiment, the housing **112** and the protrusion **114** are integrally formed. In an embodiment, the housing **112** and the protrusion **114** are separately formed and joined together to form a unitary body. It is appreciated that the housing **112** and the protrusion **114** may be formed for the same material. In an example, the material is titanium alloy.

(20) The second member **120** comprises a base **122** and a receptacle **124** extending from the base **122**. The receptacle **124** is configured to receive the protrusion **114** of the first member **110** so as to facilitate coupling of the first member **110** with the second member **120**. In an embodiment, the base **122** and the receptacle are integrally formed. In an embodiment, the base **122** and the receptacle are separately formed and joined together to form a unitary body. It is appreciated that the base **122** and the receptacle **124** may be formed for the same material. In an example, the material is titanium alloy.

(21) The receptacle **124** may have dimensions corresponding to the dimensions of the protrusion **114** to allow the protrusion **114** to be securely received within the receptacle **124**. The receptacle **124** may comprise internal threading (may not be seen in drawings) corresponding to the external threading **114**A of the protrusion **114**, the internal threading being in engagement with the external threading **114**A when the first member **110** is coupled to the second member **120** and the wearable device **100** is in the assembled state thereof.

(22) The wearable device **100** in the assembled state thereof is configured to be worn by a user. For instance, the wearable device **100** is an ear ring and a user may wear the wearable device **100** in place of a conventional car ring. The wearable device **100** in such a scenario may be designed and sized to resemble an ear ring such that a user can wear the wearable device **100**. The protrusion **114** of the first member **110** may have a length so as to pass a pierced hole in an ear (e.g., earlobe) of a user when a user wears the wearable device **100**. The base **122** of the second member **120** may comprise flanges so as to support the wearable device **100** and allow the wearable device **100** to be securely worn by the user.

(23) Reference is made to FIGS. **6-7** that illustrate the wearable device **100** being an car ring. As seen in FIG. **6**, the wearable device **100** is being placed into an ear **105** of a user, in that, the wearable device **100** is in a disassembled state with the first member **110** being inserted into the pierced hole **107** in the ear **105** from one side and the second member **120** being inserted into the pierced hole **107** in the ear **105** from an opposite side. The first member **110** may be inserted into the pierced hole **107** through the protrusion **114** that passes through the pierced hole **107** in the ear **105** of the user. As seen in FIG. **7**, the wearable device **100** has been placed into the ear **105** and the wearable device **100** is the assembled state, the first member **110** being an external facing part of the wearable device **100**. In other words, the first member **110** is an ornamental piece of the ear ring. The ornamental piece may be a stud.

(24) The housing **112** of the first member **110** may form an external facing part of the wearable device **100**, that is, the housing **112** may be visible to other users when the wearable device **100** has been worn by a user. In an embodiment, the housing **112** may be shaped to increase the aesthetic appearance of the housing **112**. In an embodiment, the housing **112** may comprise or have attached thereto external elements (e.g., ornamental elements) to increase the aesthetic appearance of the

housing **112**. The ornamental elements may be interchangeable so as to satisfy aesthetic needs of the user wearing the wearable device **100**. As described previously, the ornamental elements may comprise studs, which may be interchangeable. The user wearing the wearable device **100** may select one stud of the plurality of studs to wear at one time.

(25) The housing **112** comprises a top surface **113** forming a part of the external facing part of the wearable device **100**. As seen in FIGS. **1-2**, the top surface **113** may comprise an opening **115** formed therein, the opening **115** being in fluid communication within an internal region of the housing **112**, as will be detailed further below.

(26) Reference is now made to FIG. **3** that illustrates a cross-sectional view of the first member **110** of the wearable device **100**, the cross-section being taken along line A**1**-A**1** in FIG. **2**. The housing **112** is connected to the protrusion **114**, in that, the protrusion **114** extends from the housing **112**. The housing **112** comprises the top surface **113** forming part of the external facing surface of the first member **110** of the wearable device **100**. The housing **112** further comprises an opening **115** formed in the top surface **113**.

(27) The opening **115** is in fluid communication with an inside of the housing **112** by virtue of passage **116**. That is, the housing **112** comprises passage **116** there-within, the passage **116** extending from the opening **115** to an internal of the housing **112** and being in fluid communication with an exterior of the housing **112**, and thus, an exterior of the first member **110**. In an embodiment, the opening **115** is a circumferential opening and the passage **116** thus forms a circumferential passage. In some embodiments, the passage **116** extends into the protrusion **114** as well, in that, the protrusion **114** may comprise a part of the passage **116**, and thus, an inside of the protrusion **114** may be in fluid communication with an exterior of the housing **112** and an exterior of the first member **110**.

(28) It is appreciated that the opening **115** and the passage **116** may be of any desired configuration as long as the opening **115** facilitates the passage **116** to be in fluid communication with an exterior of the housing **112**. In some embodiments, the opening **115** may be a single opening and the passage **116** may be a single passage. In an embodiment, the opening **115** may be divided into a plurality of openings and the passage **116** may be divided into a plurality of passages corresponding to the plurality of openings. For instance, the housing **112** may comprise one or more dividers and/or ribs within the opening **115** and the passage **116**.

(29) The first member **110** of the wearable device **100** comprises a chamber **117** configured to store a liquid ink **118** therein. In an embodiment, the chamber **117** may be provided at an intersection of the housing **112** and the protrusion **114** such that the chamber **117** is partly contained within the housing **112** and partly contained within the protrusion **114**. In an embodiment, the chamber **117** is contained wholly within the housing **112**. In an embodiment, the chamber **117** is contained wholly within the protrusion **114**.

(30) The chamber **117** is in fluid communication with the passage **116**, and via the passage **116** and the opening **115**, to an exterior of the first member **110**. That is, the ink **118** contained within the chamber **117** may flow out of the first member **110** by virtue of the passage **116** and the opening **115**. The passage **116** thus extends between the opening **115** at the housing **112** and the chamber **117** within the first member **110**.

(31) In an embodiment where the chamber **117** is provided within the housing **112**, the passage **116** may extend within the housing **112**, i.e., between the opening **115** and the chamber **117** provided within the housing **112**. In an embodiment where the chamber **117** is provided within the protrusion **114**, the passage **116** may extend within the housing **112** as well as the protrusion **114**.

(32) In an embodiment, the ink **118** may be a permanent ink. In an embodiment, the ink **118** may not be easily washable and removable. In an embodiment, the ink **118** may stain an unauthorized user and the stain may remain as it is for a plurality of days.

(33) In an embodiment, the chamber **117** may comprise a sealing element **119** configured to prevent the ink **118** contained within the chamber **117** to leak into the passage **116**. The sealing element **119**

may be configured to be ruptured when a predetermined force is exerted on the sealing element **119**, for instance by the ink **118**, thereby bringing the ink into fluid communication with the passage **116**. Accordingly, the sealing element **119** may be configured to have a sealed state in which fluid communication between the chamber **117**, and the ink **118** contained therein is prevented, and a ruptured state in which fluid communication between the chamber **117**, and the ink **118** contained therein is allowed.

(34) The first member **110** comprises an ejection module **130** configured to facilitate the ink **118** in the chamber **117** to be ejected from the chamber **117** to an exterior of the first member **110** via the passage **116** and the opening **115**. In an embodiment, the ejection module **130** may be positioned within the protrusion **114**. In an embodiment, the ejection module **130** may be positioned within the housing **112**.

(35) The ejection module **130** is operatively connected to the chamber **117** so as to facilitate the ejection of ink **118** to an exterior of the first member **110**. The ejection module **130** comprises a battery **132** and an ejector **134** coupled to the battery **132**. The ejector **134** functions to eject the ink **118** from within the chamber **117** through the passage **116** and opening **115** and subsequently out of the first member **110**. In some embodiments, the battery **132** may be a rechargeable battery. In some embodiments, the battery **132** may be a non-rechargeable battery that has a cycle life of years.

(36) The ejector **134** may provide the ink **118** with sufficient energy to rupture the sealing element **119** which prevents the fluid communication between the chamber **117** and the passage **116**. That is, the ejector **134** facilitates the sealing element **119** to switch from the sealed state to the ruptured state thereof. In an embodiment, the ejector **134** may operate on the principle of pumping. In an embodiment, the ejector **134** may comprise a mechanism to provide the ink **118** with kinetic energy. In this way, in accordance with the principles of the disclosed embodiments, the wearable device **100** inhibits episodes of tampering when worn by an individual.

(37) In an embodiment, the wearable device **100** comprises a locking mechanism configured to prevent tampering of the wearable device by an unauthorized person. The wearable device **100** is thus a tamper-proof device in which the locking mechanism prevents an unauthorized person to disassemble the wearable device **100**, i.e., shift the wearable device **100** from the assembled state to the disassembled state thereof. For instance, in case the first member **110** is in threaded engagement with the second member, the locking mechanism may prevent a user to unthread the engagement between the first member **110** and the second member **120**.

(38) In an embodiment, the locking mechanism is configured to switch between a locked state and an unlocked state. In the locked state, the locking mechanism prevents a user, such as an unauthorized person, to disassemble the wearable device **100**. In the unlocked state, the locking mechanism allows a user to disassemble the wearable device **100**. In an embodiment, the locking mechanism may shift between the locked state and the unlocked state by means of a remote device under control of an authorized user, as will be detailed further below.

(39) In an embodiment, the ejection module **130** may be operatively connected to the locking mechanism. That is, the ejection module **130** may be activated when the locking mechanism is in the locked state thereof. In an unlocked state of the locking mechanism, the ejection module **130** is not activated. Accordingly, the ejection module **130** may not be activated when an authorized user is disassembling the wearable device **100**.

(40) In addition to the locking mechanism, the activation of the ejection module **130** is also based on unauthorized tampering of the wearable device **100** by an unauthorized person when the locking mechanism is in the locked state. For instance, in case an unauthorized person is tampering with the wearable device **100**, such as by trying to disassemble the wearable device when the locking mechanism is in the locked state, the ejection module **130** is activated. In an embodiment, the wearable device **100** comprises sensing means to check unauthorized tampering by an unauthorized person.

(41) The wearable device **100** further comprises a tracking module configured to transmit location of the wearable device **100** to an external device, such as a user device of an authorized user or user devices of known connections of the authorized user, and/or user devices of government and private agencies. In an embodiment, the tracking module is a GPS™ module. In an embodiment, the tracking module is positioned within the first member **110** of the wearable device **100**. In an embodiment, the tracking module is positioned within the second member **120** of the wearable device **100**. Though it is disclosed herein that the ejection module **130** and the tracking module are comprised in the first member **110** of the wearable device **100**. However, it is contemplated that either or both of the ejection module **130** or the tracking module may be included in the second member **120**, instead of the first member **110**, of the wearable device **100**.
(42) In an embodiment, the tracking module may comprise a transceiver configured to receive satellite signals and transmit data indicative of location of the wearable device **100** to an external user device, thus allowing a user associated with the external user device to track the wearable device **100**, and consequently, track a user wearing the wearable device **100**.
(43) In an embodiment, the tracking module may comprise a processing unit comprising a processor and a memory. The processor may be configured to analyze the satellite signals received by the transceiver, extract satellite data from the received signals, and determine position of the wearable device **100**, such as in terms of latitude, longitude, and/or altitude. In an embodiment, the processor may use the principle of triangulation and determine the position of the wearable device **100**.
(44) The processor is in communication with the memory which may include one or more volatile or non-volatile data storage devices or may represent a data storage function of a device. For example, memory may include a non-volatile data storage device for storing programmed instructions or data for operation of the tracking module in accordance with some embodiments of the present invention. Memory may be used to store data that is generated during communication between the satellites and the tracking module as well as during processing performed by the processor. Memory may include a non-transitory machine-readable storage medium that may be any electronic, magnetic, optical, or other physical storage device that stores executable instructions. The machine-readable storage medium may include, for example, random access memory (RAM), read-only memory (ROM), electrically-erasable programmable read-only memory (EEPROM), flash memory, a storage drive, an optical disc, or the like. The machine-readable storage medium may be encoded with executable instructions.
(45) The transceiver of the tracking module may transmit location data indicative of the position of the wearable device to an external user device, thus allowing the wearable device **100**, and consequently, a user wearing the wearable device **100** to be tracked via the external user device. In an embodiment, the tracking module transmits location data at pre-determined regular intervals. In an embodiment, the pre-determined regular interval may be a few milliseconds, a few seconds, a few minutes, and the like. In some embodiments, the tracking module may be powered by the battery **132**.
(46) Reference is now made to FIG. **4** that illustrates a system **140** comprising the wearable device **100**, the system **140** facilitating one or more external devices to communicate and control operation of one or more components of the wearable device **100**. The system **140** comprises a user device **150** and a key fob **160**, both the user device **150** and the key fob **160** being communicatively coupled to the wearable device **100**. It is appreciated that the system **140** may comprise communication networks that allows communication of the wearable device **100** with the user device **150** and the key fob **160**. The communication networks may be any type of communication network including one or more of the Internet, local area networks (LAN), wireless networks, switch or hub connections, a telephone network (e.g., a public switched telephone (PSTN) network, a cellular network, etc.), Bluetooth network, RFID network, or the like.
(47) In some embodiments, the user device **150** may be a device associated with the user wearing

the wearable device **100**. In some embodiments, the user device **150** may be a device associated with a trusted contact of the user wearing the wearable device **100**. The user device **150** may be linked to the wearable device **100** so as to allow the location of the wearable device **100** to be tracked via the user device **150**. The user device **150** may be any type of electronic device, e.g., desktop computer, laptop computer, portable or mobile device, cell phone, smartphone, tablet computer, personal digital assistant (PDA), or the like.

(48) In some embodiments, the user device **150** may comprise an application, for instance a software application, which is executable to communicate with the wearable device **100**, for instance, with the tracking module of the wearable device **100**. In some embodiments, the application may be a web application. In some embodiments, the application may be a mobile application. The mobile application may be installed on the user device **150**. In some embodiments, the user device **150** is linked to the wearable device **100** by means of the application on the user device **150**. In some embodiments, the application may instruct the user device **150** to scan a QR code associated with the wearable device **100** so as to link the user device **150** with the wearable device **100**. In some embodiments, the QR code may comprise a unique identifier of the wearable device **100**.

(49) Upon linking the wearable device with the user device **150**, the user device **150** can receive location data from the tracking module of the wearable device. Reference is made to FIG. **5** that illustrates the user device **150** in communication with the tracking module of the wearable device **100**. As seen in FIG. **5**, a location of the wearable device **100**, and thus a user wearing the wearable device **100**, is displayed on a user interface **152** of the user device **150**. In some embodiments, the location of the wearable device **100** is updated in real-time. The user interface **152** may be associated with the application running on the user device.

(50) Referring again to FIG. **4**, the key fob **160** is configured to control operation of the locking mechanism of the wearable device **100**. The key fob **160** is assigned and associated to the locking mechanism such that only the associated key fob **160** may control operation of the locking mechanism of the wearable device **100**. Accordingly, each wearable device **100** may have an associated key fob **160**. In some embodiments, each wearable device **100** may have a primary key fob and one or more backup key fobs associated therewith.

(51) In an embodiment, the key fob **160** may be associated and assigned to the wearable device **100** by a user of the wearable device **100**, such as during a set-up procedure of the wearable device. For instance, upon an initial use of the wearable device **100**, the wearable device **100** is registered and associated with the key fob **160**. In an embodiment, a QR code may be utilized to associate the key fob **160** with the wearable device **100**.

(52) As described above, the key fob **160** controls operation of the locking mechanism of the wearable device **100**. In particular, the key fob **160** controls the switching of the locking mechanism from the locked state to the unlocked state and vice versa. The key fob **160** comprises a lock button **162** and an unlock button **164**. Upon the lock button **162** being pressed by a user, a signal is transmitted to the locking mechanism causing the locking mechanism to shift to the locked state (in case the locking mechanism was previously in the unlocked state) or stay in the locked state (in case the locking mechanism was previously in the locked state). It is appreciated that the user of the key fob **160** may be the user of the wearable device **100** or a different trusted user, such as a guardian of the user wearing the wearable device **100**.

(53) Upon the unlock button **164** being pressed by a user, a signal is transmitted to the locking mechanism causing the locking mechanism to shift to the unlocked state (in case the locking mechanism was previously in the locked state) or stay in the unlocked state (in case the locking mechanism was previously in the unlocked state). It is appreciated that the locking mechanism would be operated and controlled only with signals generated by the associated key fob **160**.

(54) Accordingly, when the lock button **162** of the associated key fob **160** is pressed, the locking mechanism is in the locked state, thus preventing the wearable device **100** to be shifted from the

assembled state to the disassembled state. When the unlock button **164** of the associated key fob **160** is pressed, the locking mechanism is in the unlocked state, allowing the wearable device **100** to be disassembled, and thus being removed from the user wearing the wearable device **100**.
(55) In use, the wearable device **100** may initially be associated with the key fob **160** and the user device **150**. A user may wear the wearable device **100** by placing the wearable device on the ear of the user. When the wearable device **100** is worn by the user, the wearable device **100** is in the assembled state, with the first member **110** engaged with the second member **120**. As the user travels to different locations, the location of the wearable device **100** can be tracked on the user device **150**, for instance, via an application installed on the user device **150**.
(56) The user may utilize the key fob **160** to operate the locking mechanism of the wearable device **100**. Upon pressing the lock button **162** of the key fob **160**, the locking mechanism is in a locked state. At such a state of the locking mechanism, the wearable device **100** cannot be disassembled, i.e., the first member **110** cannot be disengaged from the second member **120** and the wearable device **100** stays in the assembled state thereof, therefore the wearable device **100** cannot be removed from the user without the use of the force. When it is desired by the user to disassemble the wearable device **100**, the user may press the unlock button **164** to shift the locking mechanism to the unlocked state.
(57) When the wearable device **100** is in the assembled state, and the locking mechanism is in the locked state, any tampering of the wearable device **100** by an unauthorized person is prevented. As an example, an unauthorized person may try to disassemble the wearable device **100** worn by the user such as to prevent tracking of the wearable device **100**. With the locking mechanism in the locked state, any such tampering by an unauthorized person is prevented.
(58) Additionally, in case an unauthorized person tries to mishandle the wearable device **100** or forcibly disassemble the wearable device **100**, the ejection module **130** may be activated. The ejection module **130** would cause the ink **118** stored in the chamber **117** to be ejected into the passage **116**, and through the opening **115**, to the exterior of the wearable device **100**. In particular, the ink **118** is ejected to the exterior of the first member **110** of the wearable device **100**. Reference is made to FIG. **8** that illustrates wearable device **100** with the ink **118** being ejected out of the first member **110** of the wearable device **100** via the opening **115** (not visible).
(59) As seen in FIG. **8**, the ink **118** flows from the chamber **117** from within the first member **110** to the exterior of the wearable device **100**. In some embodiments, the opening **115** is a circumferential opening and the ink **118** is ejected in all directions through the opening **115**, i.e., in a 360 degree pattern. As the ink **118** is ejected out of the wearable device **100**, the ink **118** may be sprayed onto an unauthorized person in vicinity of the wearable device **100**. For instance, the ink **118** may be sprayed onto an unauthorized person trying to disassemble the wearable device **100**. The ink sprayed on an unauthorized person may thus act as a silent but universal sign of the unauthorized person being a trouble maker.
(60) In some embodiments, the ink **118** may be sprayed onto the person who is wearing the wearable device **100**. For example, when the wearable device **100** is an ear ring, and someone tries to forcibly remove the wearable device **100** from the ear of the user, the ink **118** may spray onto the ear of the user, which may be indicative of that the user needs help.
(61) In some embodiments, the ejection module **130** may cause the ink **118** to spray out in a particular pattern (e.g., when the device **100** is being forcibly removed from a person wearing it), the particular pattern being a sign of danger, help, or emergency.
(62) Such a silent sign is extremely helpful in cases of human trafficking and child abductions where a user may be forcibly taken to different locations. The wearable device would provide a two-fold advantage in that the tamper-proof device would transmit the live location to a remote user device for tracking, and further, any forced tampering would cause injection of the ink onto the unauthorized person. The ink stains on the unauthorized person would be readily visible and it would be easier for officials and general public to detect and catch the unauthorized person.

(63) Since many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Thus, the scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. A wearable device comprising: a first member having a circumferential opening and a second member configured to be detachably engaged with the first member through a threaded engagement and unthreaded engagement between the first member and the second member; a chamber comprising liquid ink, the liquid ink being in fluid communication with the circumferential opening via a passage extending between the chamber and the circumferential opening; a locking mechanism configured to switch between a locked state in the threaded engagement of the first member with the second member and an unlocked state in the unthreaded engagement of the first member with the second member; a sealing element capable of having a sealed state and a ruptured state and configured to prevent the liquid ink contained within the chamber from leaking to the passage when in the sealed state, wherein the sealing element is configured to rupture when a predetermined force is exerted thereon while in the sealed state thereby switching to the ruptured state; an ejection module positioned within either the first member or the second member and comprising an ejector and a battery, the ejection module operatively connected to the locking mechanism and comprising an ejector operatively coupled to the battery and configured to eject the liquid ink through the circumferential opening of the first member when the locking mechanism is at the locked state thereof and wherein the ejector is configured to provide the liquid ink with sufficient energy to rupture the sealing element thereby causing a transitioning of the sealing element from the sealed state to the ruptured state and ejecting the liquid ink in all directions in a 360 degree pattern such that the ejection module is activated when the locking mechanism is in the locked state thereof and deactivated when the locking mechanism is in the unlocked state thereof; and wherein the wearable device inhibits episodes of tampering when worn by an individual.
2. The wearable device of claim 1, wherein the tracking module further comprises: a transceiver configured to receive satellite signals and transmit location data indicative of the location of the wearable device; and a processing unit configured for analyzing the satellite signals received by the transceiver and determining the location of the wearable device.
3. A wearable device comprising: a first member having a circumferential opening and a second member configured to be detachably engaged with the first member through a threaded engagement and unthreaded engagement between the first member and the second member; a chamber comprising liquid ink, the liquid ink being in fluid communication with the circumferential opening via a passage extending between the chamber and the circumferential opening; a locking mechanism configured to switch between a locked state in the threaded engagement of the first member with the second member and an unlocked state in the unthreaded engagement of the first member with the second member; a sealing element capable of having a sealed state in which fluid communication between the chamber and the liquid ink contained therein is prevented, and a ruptured state in which fluid communication between the chamber and the liquid ink contained therein is allowed, and wherein the sealing element is configured to rupture when a predetermined force is exerted thereon while in the sealed state thereby transitioning to the ruptured state; an ejection module comprising an ejector configured to provide the liquid ink with sufficient energy to rupture the sealing element thereby causing the transitioning from the sealed state to the ruptured state and to eject the liquid ink in all directions in a 360 degree pattern through the circumferential opening of the first member when the locking mechanism is at the locked state

thereof; and wherein the wearable device inhibits episodes of tampering when worn by an individual.

4. The wearable device of claim 3, wherein the ejection module is operatively connected to the locking mechanism.

5. The wearable device of claim 4, wherein the ejection module is activated when the locking mechanism is in the locked state thereof and deactivated when the locking mechanism is in the unlocked state thereof.

6. The wearable device of claim 3, wherein the wearable device further comprises: a tracking module configured to transmit a location of the wearable device.

7. The wearable device of claim 6, wherein the tracking module is positioned within either the first member or the second member.

8. The wearable device of claim 3, wherein the ejection module is positioned within either the first member or the second member.

9. The wearable device of claim 6, wherein the tracking module is a Global Positioning System (GPS) module.

10. The wearable device of claim 6, wherein the tracking module further comprises: a transceiver configured to receive satellite signals and transmit location data indicative of the location of the wearable device; and a processing unit configured for analyzing the satellite signals received by the transceiver and determining the location of the wearable device.

11. The wearable device of claim 3, wherein the liquid ink is a permanent ink.

12. The wearable device of claim 3, wherein the sufficient energy is in a kinetic energy form.

13. The wearable device of claim 3, wherein the ejection module further comprises: a battery operatively coupled to the ejector, wherein the ejector is configured to provide the liquid ink with the sufficient energy to rupture the sealing element thereby causing a transitioning of the sealing element from the sealed state to the ruptured state.

14. The wearable device of claim 3, wherein the circumferential opening is divided into a plurality of openings and the passage is divided into a plurality of passages corresponding to the plurality of openings.

15. The wearable device of claim 3, wherein the wearable device is an ear ring.

16. A wearable device comprising: a first member having a circumferential opening and a second member configured to be detachably engaged with the first member through a threaded engagement and unthreaded engagement between the first member and the second member; a chamber comprising liquid ink, the liquid ink being in fluid communication with the circumferential opening via a passage extending between the chamber and the circumferential opening; a locking mechanism configured to switch between a locked state in the threaded engagement of the first member with the second member and an unlocked state in the unthreaded engagement of the first member with the second member; a sealing element capable of having a sealed state in which fluid communication between the chamber and the liquid ink contained therein is prevented, and a ruptured state in which fluid communication between the chamber and the liquid ink contained therein is allowed, and wherein the sealing element is configured to rupture when a predetermined force is exerted thereon while in the sealed state thereby transitioning to the ruptured state; an ejection module operatively connected to the locking mechanism and comprising an ejector configured to provide the liquid ink with sufficient energy to rupture the sealing element thereby causing the transitioning from the sealed state to the ruptured state and to eject the liquid ink in all directions in a 360 degree pattern through the circumferential opening of the first member when the locking mechanism is at the locked state thereof; and wherein the wearable device inhibits episodes of tampering when worn by an individual.

17. The wearable device of claim 16, wherein the ejection module further comprises: a battery operatively coupled to the ejector coupled to the battery, wherein the ejector is configured to provide the liquid ink with the sufficient energy to rupture the sealing element thereby causing a

transitioning of the sealing element from the sealed state to the ruptured state such that the ejection module is activated when the locking mechanism is in the locked state thereof and deactivated when the locking mechanism is in the unlocked state thereof.

18. The wearable device of claim 16, wherein the circumferential opening is divided into a plurality of openings and the passage is divided into a plurality of passages corresponding to the plurality of openings.